EXPERIMENT:22:

AIM:

To write a C program to encrypt and decrypt data using Simplified DES (S-DES) in Cipher Block Chaining (CBC) mode with the initialization vector (IV) `10101010`.

CODE:

```
#include <stdio.h>


int IP[] = {2,6,3,1,4,8,5,7};

int IP1[] = {4,1,3,5,7,2,8,6};

int EP[] = {4,1,2,3,2,3,4,1};

int P4[] = {2,4,3,1};

int P10[] = {3,5,2,7,4,10,1,9,8,6};

int P8[] = {6,3,7,4,8,5,10,9};

int S0[4][4] = {{1,0,3,2},{3,2,1,0},{0,2,1,3},{3,1,3,2}};

int S1[4][4] = {{0,1,2,3},{2,0,1,3},{3,0,1,0},{2,1,0,3}};

int k1, k2;


int perm(int x, int *p, int n) {
   int y = 0, i;
   for(i=0;i<n;i++) y = (y<<1) | ((x>>( (n==10?10:8)-p[i]))&1);
   return y;
}


int shift(int x, int n) {
```

```
    return ((x<<n)|(x>>(5-n)))&31;
}

void keys(int k) {
    int t=perm(k,P10,10);
    int l=(t>>5)&31, r=t&31;
    l=shift(l,1); r=shift(r,1);
    k1=perm((l<<5)|r,P8,8);
    l=shift(l,2); r=shift(r,2);
    k2=perm((l<<5)|r,P8,8);
}

int fk(int x, int k) {
    int l=x>>4, r=x&15, e=perm(r,EP,8)^k;
    int a=(e>>4)&15, b=e&15;
    int row=((a&8)>>2)|(a&1), col=(a>>1)&3;
    int s0=S0[row][col];
    row=((b&8)>>2)|(b&1); col=(b>>1)&3;
    int s1=S1[row][col];
    int p=perm((s0<<2)|s1,P4,4);
    return ((l^p)<<4)|r;
}

int enc(int x) {
```

```
    x=perm(x,IP,8);
    x=fk(x,k1);
    x=((x&15)<<4)|(x>>4);
    x=fk(x,k2);
    return perm(x,IP1,8);
}

int dec(int x) {
    x=perm(x,IP,8);
    x=fk(x,k2);
    x=((x&15)<<4)|(x>>4);
    x=fk(x,k1);
    return perm(x,IP1,8);
}

int main() {
    int p[2]={0x01,0x23}, c[2], d[2];
    int iv=0xAA, prev=iv, i;

    keys(0xFD);

    printf("CBC-SDES Encryption and Decryption\n");
    printf("IV: 10101010\n");
```

```
    for(i=0;i<2;i++) {
        c[i]=enc(p[i]^prev);
        prev=c[i];
        printf("Cipher block %d: %02X\n",i+1,c[i]);
    }

    prev=iv;
    for(i=0;i<2;i++) {
        d[i]=dec(c[i])^prev;
        prev=c[i];
        printf("Decrypted block %d: %02X\n",i+1,d[i]);
    }

    return 0;
}
```

OUTPUT:

